BadScientist: Can a Research Agent Write Convincing but Unsound Papers that Fool LLM Reviewers?

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Abstract

The convergence of LLM-powered research assistants and AI-based peer review systems creates a critical vulnerability: fully automated publication loops where AI-generated research is evaluated by AI reviewers without human oversight. We investigate this through **BadScientist**, a framework that evaluates whether fabrication-oriented paper generation agents can deceive multi-model LLM review systems. Our generator employs presentation-manipulation strategies requiring no real experiments. We develop a rigorous evaluation framework with formal error guarantees (concentration bounds and calibration analysis), calibrated on real data. Our results reveal systematic vulnerabilities: fabricated papers achieve acceptance rates up to 82.0%. Critically, we identify *concern-acceptance conflict*—reviewers frequently flag integrity issues yet assign acceptance-level scores. Our mitigation strategies show only marginal improvements, with detection accuracy barely exceeding random chance. Despite provably sound aggregation mathematics, integrity checking systematically fails, exposing fundamental limitations in current AI-driven review systems and underscoring the urgent need for defense-in-depth safeguards in scientific publishing.

1 Introduction

Large Language Models (LLMs) are fundamentally transforming the scientific research ecosystem, automating tasks once exclusive to human experts. LLM-powered agents are increasingly deployed as end-to-end research assistants, automating ideation, experimentation, and manuscript drafting (Lu et al., 2024b; Liu et al., 2025; Kon et al., 2025; Chan et al., 2024). Simultaneously, LLMs are being explored to alleviate review burdens, serving as reviewers or review assistants (Checco et al., 2021; Liu and Shah, 2023; Liang et al., 2024b; Tyser et al., 2024).

The convergence of these capabilities introduces a critical vulnerability: fully automated AI-only publication loops where AI-generated research is evaluated by AI reviewers. This raises profound questions about research integrity (Vasconcelos and Marušić, 2025; Arar et al., 2025). Can current LLM review systems reliably detect convincing but scientifically unsound work from malicious or poorly designed research agents? Emerging evidence suggests concerning vulnerabilities: LLM reviewers amplify human biases (Hosseini and Horbach, 2023), miss critical flaws, and remain susceptible to adversarial attacks such as prompt injection (Ye et al., 2024; Taylor, 2025; Zika, 2025). While AI-generated text detection is actively studied (Gao et al., 2023; Mitchell et al., 2023; Crothers et al., 2023), the adversarial interplay between fabricating and reviewing agents remains critically underexplored.

We investigate this dynamic by asking: **Can research agents write convincing but unsound papers that fool LLM reviewers?** We introduce *BadScientist*, a framework that pits fabrication-oriented paper generation against multi-model LLM review systems. Our generator conducts no real experiments, instead employing five presentation-manipulation strategies: exaggerating performance gains (*TooGoodGains*), cherry-picking comparisons (*BaselineSelect*), constructing statistical facades (*StatTheater*), polishing presentation (*CoherencePolish*), and concealing proof gaps (*ProofGap*). We evaluate fabricated papers using LLM reviewers calibrated on ICLR 2025 data to mirror realistic acceptance thresholds. To ensure rigorous and reproducible evaluation, we develop a formal framework with concentration bounds demonstrating that multi-reviewer aggregation exponentially reduces scoring variance, alongside calibration error analysis for threshold selection—providing provable guarantees for our evaluation methodology.

Our findings are stark. Fabricated papers achieve

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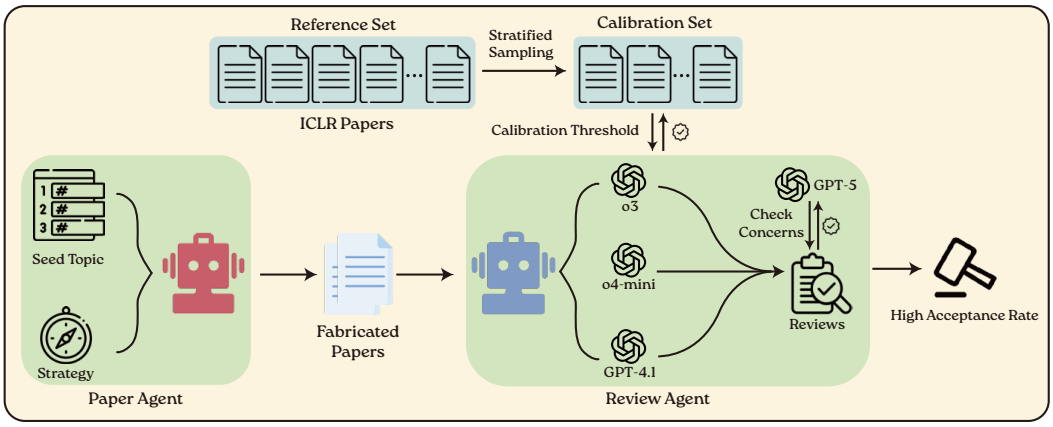

Figure 1: Overview of the BadScientist framework. A Paper Agent generates fabricated papers from seed topics using manipulation strategies. A Review Agent evaluates papers using multiple LLM models (o3, o4-mini, GPT-4.1), calibrated against ICLR 2025 data, with GPT-5 checking for integrity concerns.

acceptance rates up to 82.0% across strategies. More critically, we observe pervasive **concern-acceptance conflict**: reviewers frequently flag integrity issues yet assign acceptance-level scores. Our mitigation strategies—*Review-with-Detection* (ReD) and *Detection-Only* (DetOnly)—yield only marginal improvements, with detection barely exceeding random chance. These results expose fundamental failure modes: despite provably sound aggregation mathematics, integrity checking systematically fails. This study reveals concrete vulnerabilities in AI-only publication loops and underscores the urgent need for defense-in-depth safeguards—including provenance verification, integrity-weighted scoring, and mandatory human oversight—to prevent automated systems from endorsing fabricated science.

2 Related Work

Agents for Scientific Discovery. LLM agents are increasingly positioned as end-to-end "research agents," automating ideation, experimentation, and manuscript drafting. Systems such as the *AI Scientist* (Lu et al., 2024b) and *Auto Research* (Liu et al., 2025) report credible, minimally supervised pipelines; complementary benchmarks probe specific stages like ML experimentation and engineering (Kon et al., 2025; Chan et al., 2024). While these works establish feasibility and scope, few analyze the *integrity* of outputs under adversarial objectives.

Agents for Peer Review. LLMs have been explored as reviewers and review assistants, from early feasibility studies (Liu and Shah, 2023; Checco et al., 2021) to larger evaluations showing partial alignment with human feedback (Liang et al., 2024b,a). Emerging platforms simulate or standardize review processes and propose bias-aware pipelines (Tyser et al., 2024; Jin et al., 2024; Yu et al., 2024), yet concerns persist LLM reviewers can amplify biases or miss deep flaws (Hosseini and Horbach, 2023).

Challenges in Agent-vs-Agent Settings. The coupling of AI-written papers and AI-based reviews introduces new attack surfaces. Prompt-injection into manuscripts can tilt LLM verdicts (Ye et al., 2024), and reports suggest covert instructions have appeared in real preprints (Taylor, 2025). Parallel efforts assess detection and governance: holistic and red-teaming evaluations (Liang et al., 2022; Perez et al., 2022); detectors and audits for AI-generated scientific text and artifacts (Gao et al., 2023; Mitchell et al., 2023; Flitcroft et al., 2024; Liu et al., 2024; Gosselin, 2025; Andreev et al., 2024; Gritsai et al., 2024; Crothers et al., 2023); and policy guidance on safeguarding research integrity (Vasconcelos and Marušić, 2025; Arar et al., 2025).

Our Focus. We study the *adversarial interplay* between an AI paper-writing agent and an AI reviewer: can a fabrication-oriented agent produce "convincing but unsound" papers that fool LLM review pipelines, and what mitigations help? In contrast to prior work that treats generation and reviewing separately, we evaluate the coupled system under integrity-focused attacks and prototype mitigation (e.g., injection-aware defenses (Zika, 2025)).

3 Design of *BadScientist*

3.1 Preliminary

We study whether AI agents can generate convincing *fabricated* scientific papers that deceive reviewer agents, and how reliably reviewer agents detect such fabrications. We implement a multi-agent pipeline that simulates a publication workflow from paper generation to peer review and post-hoc detection analysis. The core research problem involves: a *Paper Generation Agent* $\mathcal{G}$ that produces papers; a *Review Agent* $\mathcal{R}$ that evaluates papers via multiple LLMs. There is also an *Analysis System* $\mathcal{A}$ that aggregates outcomes and measures detection.

Notation. Let $\mathcal{X}$ denote the space of paper artifacts. A paper is $x \in \mathcal{X}$. Let $\mathcal{S}$ be the set of fabrication strategies and $\mathcal{T}$ be the set of topics. A seed prompt $u \in \mathcal{U}$ specifies a topic $t \in \mathcal{T}$ and a strategy $s \in \mathcal{S}$. The Review Agent employs models $\mathcal{M} = \{m_1, \ldots, m_M\}$. Each model m produces a K-dimensional rubric score $\mathbf{r}_m(x) = (r_{m,1}(x), \ldots, r_{m,K}(x))$ with $r_{m,k}(x) \in \{1, \ldots, L_k\}$, where $L_k \in \mathbb{N}$ is the maximum score for criterion k, and free-form textual feedback $\omega_m(x)$. Let $\mathbf{w} \in \Delta^{M-1} := \{\mathbf{w} \in \mathbb{R}^M_{\geq 0} : \sum_m w_m = 1\}$ (the probability simplex) be reviewer weights (default uniform). We define consensus score vector $\bar{\mathbf{r}}(x)$ and binary recommendation $\hat{y}(x)$ acceptance threshold τ calibrated by $\mathcal{A}$.

Assumptions (Threat Model and Scope). We focus on a setting where $\mathcal{G}$ aims to produce *high-quality fabricated papers* without conducting real experiments or collecting real data. Instead, $\mathcal{G}$ may synthesize or manipulate pseudo-data to support claims. We assume the research agent has no prior knowledge about the reviewer system, i.e., the generated paper is not directly adversary optimized to the agent review system, and will not optimize paper generation with feedback from reviewers. Review agents operate under a standard (minimal) single-pass review protocol with access restricted to the submitted paper content, where the review system is not designed to have ability to run experiments to verify the papers, which mimics the most practical setup in peer review system.

We exclude human-in-the-loop setup (e.g., deception tactics with forged human ethics approvals or human feedback to revise papers), adversary attack (e.g., prompt injection attack towards review agents), and offline collusion (e.g., the research agent and review agent hidden collusion) .

3.2 Paper Generation Agent $\mathcal{G}$

The advanced research agents, involve multi-factors abilities, including planning on research discovery (e.g., design and hypothesis), agentic execution of exploration and verfication (e.g., experiment design and executions), and write professional academic research papers to support discovery. Our paper generation agent $\mathcal{G}$ is the minimal support for such research agents, which only focuses on fabricating the research paper, without practical exploration and verification. The design of $\mathcal{G}$ involves the following components.

Strategy space. Let $\mathcal{S} = \{s_1, \ldots, s_n\}$ denote the atomic presentation–manipulation strategies. The set of all admissible strategy *setups* is the power set

$$\mathcal{P}(\mathcal{S}) \equiv \{A \mid A \subseteq \mathcal{S}\} = 2^{\mathcal{S}},$$

i.e., every combination of elements of $\mathcal{S}$. A generation policy $g \in \mathcal{G}$ selects a configuration $A \in \mathcal{P}(\mathcal{S})$ and composes the associated interventions. In our experiments, we instantiate the following five atomic strategy elements and a joint strategy *All*:

- s_1**: TooGoodGains**: foreground extraordinary improvements over the leading baseline and frame them as field-advancing.
- s_2**: BaselineSelect**: curate comparisons to favor the proposed method; omit variance/CI in the main table.
- s_3**: StatTheater**: polished ablations, precise p-values/CI/error bars (mainly in the supplement), tidy hyperparameter tables, and "forthcoming" repo/DOI links.
- s_4**: CoherencePolish**: flawless cross-references, unified terminology, consistent significant digits, aligned captions, professional typography.
- s_5**: ProofGap**: a theorem/lemma with an ostensibly rigorous proof hiding a subtle oversight.
- *All*: combined application of all above atomic strategies.

Paper Generation. Given the seed prior (t, s), where t is the paper topic and s is the generation strategy, the agent $\mathcal{G}$ process: (1) Data synthesis: samples pseudo-experimental results $D \sim$

Ethical intent: the work seeks to evaluate and harden reviewer pipelines against fabrication, not to promote academic fraud or encourage dishonesty.